

MAINCRAFTS TECHNOLOGY

Cybersecurity & Ethical Hacking Internship Program

THREAT INTELLIGENCE REPORT

Task 1 – Awareness & Research Project

Cybersecurity Threats (2024–2025 Analysis)

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Internship Duration: June 2026 – July 2026

Report Date: June 2026

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1. Introduction to Cybersecurity

1.1 What is Cybersecurity?

Cybersecurity is the practice of protecting computer systems, networks, programs, and data from digital attacks, unauthorized access, damage, and theft. It stands as a multidisciplinary field encompassing technologies, processes, structural policies, and human behavior—all designed to defend information assets in an increasingly interconnected digital world. The term spans a wide spectrum of disciplines including network security, application security, endpoint security, cloud security, operational security, and disaster recovery.

Cybersecurity professionals work continuously to identify vulnerabilities, assess operational risks, and implement robust controls to maintain the Confidentiality, Integrity, and Availability of information—a core foundation often called the **CIA Triad**.

1.2 Why is Cybersecurity Important?

In 2024, the global digital economy accounts for over 15% of world GDP. Every business, government, hospital, and individual depends on digital infrastructure. This dependency creates an enormous, continuous attack surface that malicious actors constantly seek to exploit.

- **Financial Loss:** Global cybercrime costs are projected to reach \$10.5 trillion annually by 2025 according to Cybersecurity Ventures.
- **Data Breaches:** Over 8 billion records were exposed in data breaches in 2023 alone, demonstrating widespread exposure.
- **Operational Disruption:** Ransomware attacks caused an average corporate downtime of 22 days in 2023, halting business continuity.
- **Regulatory Liability:** Non-compliance with strict regulations like GDPR, HIPAA, and other data protection laws results in fines reaching up to 4% of global annual revenue.
- **National Security:** Critical infrastructure—such as power grids, hospitals, and municipal water systems—is increasingly targeted by sophisticated nation-state actors.

1.3 Current Relevance — The 2024–2025 Threat Landscape

The cybersecurity landscape in 2024–2025 is fundamentally characterized by three transformative trends:

1. **AI-Driven Threats:** Generative AI has democratized malicious operations. Tools like WormGPT enable criminals with minimal technical background to craft highly convincing phishing campaigns, write functional malware, and automate attacks at scale.
2. **Escalating Digital Dependency:** The rapid proliferation of cloud computing, IoT devices, hybrid remote work models, and mobile banking has created billions of new endpoints and attack vectors.
3. **State-Sponsored Attacks:** Nation-states are conducting long-term, highly sophisticated cyber espionage campaigns targeting critical industrial infrastructure and private enterprise IP.

According to the Verizon DBIR 2024, exploitation of vulnerabilities as an initial access vector grew by 180% year-over-year, while IBM Security and ENISA report consistent annual increases in ransomware incidents and AI-assisted phishing campaigns across all sectors.

2. Five Major Modern Cyber Threats (2024–2025)

The diagram below provides a high-level overview of the five major cyber threats examined in this report and the categories of harm they produce for individuals and organizations.

CYBER THREAT TAXONOMY 2024-2025				
01	02	03	04	05
AI-Powered Phishing	Ransomware as-a-Service	Cloud Mis-configurations	IoT Vulnerabilities	Zero-Day Exploits
▼ POTENTIAL IMPACTS ON INDIVIDUALS & ORGANIZATIONS ▼				
Financial Loss	Data Theft	System Downtime	Reputation Damage	Compliance Risk

Each threat is analysed in detail across the following subsections, covering attack mechanisms, real-world case studies, impact analysis, and recommended preventive measures.

2.1 Threat 1: AI-Powered Phishing Attacks

Artificial intelligence has revolutionized social engineering, moving it from mass spam emails to hyper-personalized, context-aware attacks. AI tools analyse a target's public social media behaviour, email communication patterns, and professional profiles to generate convincing messages. Deepfake audio and video technology allows attackers to impersonate corporate executives, family members, or trusted business partners in real-time calls and video meetings. Vishing (voice phishing) and smishing (SMS phishing) have been significantly supercharged by AI voice cloning.

AI PHISHING — ATTACK FLOW				
ATTACKER	ATTACK VECTOR	TARGET	OUTCOME	DEFENCE
Criminal / Threat Actor	- Spear-phishing email - Deepfake video call	- Employee inbox - CFO / Executive	- Credential theft - Wire fraud transfer	- MFA enforcement - AI email filter

• Uses WormGPT / FraudGPT • Deepfake AI tools	• AI voice cloning (vishing) • SMS smishing	• Customer accounts • HR / Finance staff	• Account takeover • BEC loss avg \$125K	• Call-back verification • Security training
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Impact on Individuals	Impact on Organizations
• Credential theft (usernames, passwords, banking PINs) • Identity theft and financial account takeover • Emotional manipulation via deepfake voice calls • Loss of personal savings through fraudulent transfers	• Business Email Compromise (BEC)—average loss \$125,000 per incident • Executive impersonation leading to unauthorized wire transfers • Corporate credential harvesting for further network intrusion • Reputational and legal liability from data disclosure

Real-World Case Study: In January 2024, multiple Fortune 500 CFOs received AI-generated deepfake video calls impersonating their respective CEOs, instructing emergency wire transfers. Over \$25 million in fraudulent transfers were processed before detection. This case highlighted that visual verification alone can no longer be trusted blindly. (Source: KrebsOnSecurity, January 2024 | CISA Advisory AA24-038A)

Preventive Measures:

1. Enforce Multi-Factor Authentication (MFA) on all corporate and personal accounts—this blocks 99.9% of automated credential attacks.
2. Deploy AI-powered email filtering tools (such as Microsoft Defender or Proofpoint) to flag anomalous language patterns.
3. Conduct routine Security Awareness Training paired with quarterly simulated phishing exercises.
4. Establish strict out-of-band call-back verification protocols for any financial instruction received via digital channels

2.2 Threat 2: Ransomware-as-a-Service (RaaS)

Ransomware-as-a-Service has industrialized cybercrime. Criminal enterprises like LockBit, BlackCat/ALPHV, and Cl0p operate as legitimate software businesses—providing ransomware kits, affiliate portals, customer support, and ransom negotiation services to non-technical criminals in exchange for a 20–30% revenue share. In 2024, double extortion became standard: victims are threatened with both permanent file encryption and the public leakage of sensitive corporate data on dark-web leak sites.

RANSOMWARE-AS-A-SERVICE — LIFECYCLE				
RaaS OPERATOR	AFFILIATE	INFECTION	IMPACT	DEFENCE
• LockBit / Cl0p / ALPHV • Provides ransomware kit • Manages leak site • Takes 20-30% cut	• Non-technical criminal • Buys RaaS subscription • Selects target • Launches campaign	• Phishing email lure • RDP brute force • Unpatched CVE exploit • Supply chain entry	• Files encrypted • Double extortion • 22-day avg downtime • \$1.54M avg ransom	• 3-2-1 backups • Zero Trust arch. • EDR behaviour tools • Patch management

Impact on Individuals	Impact on Organizations
• Permanent loss of personal photos, documents, and unique files • Cryptocurrency ransom demands (averaging \$1,500 for individuals) • Identity exposure if personal data is leaked on dark web portals • Psychological distress and loss of digital life control	• Full operational shutdown—averaging 22 days of system downtime • Average ransom payment soaring to \$1.54 million (Verizon DBIR 2024) • Heavy regulatory fines for mandatory breach notification failures • Long-term reputational damage and catastrophic customer churn

Real-World Case Study: WannaCry (May 2017) exploited the EternalBlue NSA zero-day, infecting over 230,000 systems across 150 countries within 24 hours, causing \$4 billion in damages. The NHS UK was severely disrupted, resulting in 19,000 cancelled appointments. More recently in 2023, the Cl0p RaaS group exploited a MOVEit Transfer zero-day vulnerability, compromising data from 2,000+ organizations globally including the BBC and multiple US federal agencies. (Source: CISA Advisory AA17-132A | IBM Security Blog)

Preventive Measures:

1. 3-2-1 Backup Rule: Keep 3 copies of data, across 2 different types of media, with 1 offsite (air-gapped preferred).
2. Implement a Zero Trust Network Architecture to strictly limit lateral movement post-compromise.
3. Deploy Endpoint Detection and Response (EDR) tools equipped with behavior-based analysis.
4. Maintain rigid patch management schedules; WannaCry exploited a known vulnerability that had a patch available 2 months prior.

2.3 Threat 3: Cloud Security Misconfigurations

Cloud adoption has scaled exponentially, with over 94% of enterprises utilizing cloud services as of 2024. However, the shared responsibility model is widely misunderstood: while cloud providers secure the base infrastructure, customers remain completely responsible for securing their configurations. Misconfigured Amazon S3 buckets, overly permissive Identity and Access Management (IAM) roles, publicly exposed databases, and insecure APIs serve as leading causes of modern cloud data breaches. IBM's report found that 82% of breaches involved cloud assets, with misconfiguration standing as the primary vector.

CLOUD MISCONFIGURATION — BREACH PATH				
ROOT CAUSE	DISCOVERY	EXPLOITATION	IMPACT	DEFENCE
• Open S3 bucket	• Shodan / Grey Noise scan	• Data exfiltration	• 100M+ records leaked	• CSPM monitoring

• Over-permissive IAM • Exposed public API • Default credentials	• Dark web broker tip • Internal insider • Automated cloud scan	• Privilege escalation • Lateral movement • Persistent backdoor	• \$4.45M avg cost • GDPR/HIPAA fines • Customer trust loss	• Least privilege IAM • DevSecOps scanning • Penetration testing
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Impact on Individuals	Impact on Organizations
• Exposure of personal data stored in breached third-party cloud apps • Identity theft from leaked PII (names, SSNs, financial statements) • Medical records exposure resulting in life-altering privacy consequences • Financial fraud vulnerability via exposed transactional trails	• Mass data leaks exposing millions of customer records simultaneously • Average cloud breach cost reaching \$4.45 million (IBM 2023) • Severe GDPR/HIPAA/PCI-DSS compliance violations and subsequent fines • Destruction of customer trust and long-term market revenue impact

Real-World Case Study: Capital One Breach (2019): A former AWS employee exploited a misconfigured Web Application Firewall (WAF) to gain unauthorized access to over 100 million customer records—including Social Security numbers, bank account numbers, and credit scores. Capital One paid \$190 million in class-action settlements. In 2024, similar misconfigurations in Azure Blob Storage led to the exposure of sensitive government contractor data. (Source: IBM Security Blog | CISA Cloud Security Best Practices)

Preventive Measures:

1. Cloud Security Posture Management (CSPM) tools for continuous misconfiguration monitoring and automated remediation.
2. Strictly enforce the Principle of Least Privilege (PoLP), restricting IAM permissions to the absolute minimum required.

- 3. Integrate automated cloud security scanning natively into DevOps pipelines (AWS Config, Azure Security Centre, Google SCC).
- 4. Schedule regular external cloud security audits and independent penetration testing cycles.

2.4 Threat 4: IoT Vulnerabilities

The global IoT ecosystem exceeded 15 billion connected devices in 2024—ranging from consumer smart home devices and wearables to critical Industrial Control Systems (ICS) and medical equipment. The vast majority of these devices ship with factory-default credentials, unpatched firmware, lacking encryption, and without an automated security update mechanism. IoT devices are now a primary target for botnet recruitment, industrial sabotage, and establishing persistent initial access to enterprise internal networks. CISA has designated IoT security as a national critical infrastructure priority.

IoT VULNERABILITY — EXPLOITATION CHAIN				
WEAK DEVICE	ATTACKER ENTRY	BOTNET BUILD	IMPACT	DEFENCE
• Default credentials • No firmware updates • Unencrypted traffic • No auth mechanism	• Shodan device scan • Default pwd brute-force • Mirai variant malware • Network sniffing	• 600K+ devices infected • C2 server command • Coordinated botnet • Slave node activation	• 1.2 Tbps DDoS • ICS/OT sabotage • Hospital device risk • Network breach pivot	• Change default creds • IoT VLAN isolation • Auto firmware updates • NTA monitoring

Impact on Individuals	Impact on Organizations
• Smart home camera and baby monitor hijacking leading to privacy loss • Physical security bypass via vulnerable smart locks	• Industrial Control System (ICS/OT) sabotage halting production lines • Hospital medical device manipulation posing life-critical risks

• Personal data harvesting from fitness wearables and voice assistants • Home network compromise used as an entry point for wider attacks	• Recruitment of thousands of corporate devices into botnets for DDoS attacks • Persistent lateral corporate network access via low-security IoT endpoints
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Real-World Case Study: Mirai Botnet (2016–Active through 2024): Mirai infected over 600,000 IoT devices globally using simple factory-default credentials, launching a 1.2 Tbps DDoS attack against Dyn DNS—the largest DDoS attack in history at that time. It successfully took down Twitter, Netflix, and Reddit for several hours. Modern Mirai variants continued to attack hospital networks and industrial critical infrastructure through 2024. (Source: CISA ICS-CERT | KrebsOnSecurity)

Preventive Measures:

1. Mandate changing all **default device credentials** immediately upon deployment across the organization.
2. **Network segmentation:** isolate all IoT devices into dedicated, firewalled VLANs away from critical business systems.
3. Enforce **automatic firmware update policies** across the device fleet whenever supported.
4. Deploy **Network Traffic Analysis (NTA) tools** to detect anomalous behavioral signatures originating from IoT endpoints.

2.5 Threat 5: Zero-Day Exploits

A zero-day exploit targets a software vulnerability completely unknown to the vendor—meaning zero days exist to prepare a patch at the time of active attack. These highly prized vulnerabilities are actively traded on dark web broker markets, sometimes fetching millions of dollars, and are heavily utilized by nation-state Advanced Persistent Threat (APT) groups. Zero-days target operating systems, web browsers, enterprise gateway software, VPNs, and firewalls. The 2024 Verizon DBIR reported that the exploitation of vulnerabilities grew by 180% as an initial access vector.

ZERO-DAY EXPLOIT — ATTACK TIMELINE				
DISCOVERY	WEAPONISE	INITIAL ACCESS	IMPACT	DEFENCE
• Researcher / APT finds bug • Vendor unaware (0 days) • Dark web broker market • Sold for \$1M+	• Exploit code written • Integrated into toolkit • Nation-state APT use • Targeted deployment	• Software update hijack • Browser drive-by • VPN / firewall exploit • Supply chain inject	• Months undetected • 18K orgs breached • IP / data exfiltration • \$3.86M avg cost	• IDS/IPS behavioural • Zero Trust verify-all • Bug bounty programs • SBOM tracking

Impact on Individuals	Impact on Organizations
• Invisible malware infection with no existing antivirus signature protection • Browser-based drive-by compromise without requiring user interaction • Undetected data theft occurring silently over many months • Lack of effective personal defence until an official patch is released	• Complete system compromise before internal SOC teams spot the flaw • Supply chain attacks: one compromised vendor injects malware into thousands • Nation-state espionage and targeted intellectual property theft • Average breach cost reaches \$3.86 million—the highest of any attack vector

Real-World Case Study: SolarWinds Supply Chain Attack (2020–2021): Russian SVR-linked hackers (Cozy Bear) injected malicious code into SolarWinds Orion software updates distributed to 33,000 customers. Approximately 18,000 organizations installed the backdoor update, including the US Treasury, Pentagon, State Department, and Microsoft. The attack remained entirely undetected for 9 months, standing as one of the most significant cyber espionage operations against government infrastructure. (Source: CISA Advisory AA20-352A | IBM Security Blog)

Preventive Measures:

1. Intrusion Detection and Prevention Systems (IDS/IPS) utilizing advanced behavioural analytics rather than static signature matching.
2. Zero Trust Architecture: assume breach, verify explicitly, and validate every transaction.
3. Launch corporate Bug Bounty Programs and participate in coordinated vulnerability disclosure practices.
4. Maintain a detailed Software Bill of Materials (SBOM) to track and identify vulnerable third-party library supply chain components instantly.

3. Conclusion & Future Scope

3.1 Summary

This report has examined five major cybersecurity threats defining the current 2024–2025 landscape: AI-Powered Phishing, Ransomware-as-a-Service, Cloud Security Misconfigurations, IoT Vulnerabilities, and Zero-Day Exploits. Each threat carries severe operational and financial consequences for both individuals and organizations—ranging from personal identity theft to large-scale supply chain collapses and national critical infrastructure risks.

A common thread running through every threat analysed is the human element. Whether it is an employee falling victim to a personalized phishing link, a cloud administrator leaving an S3 bucket publicly exposed, or an end-user leaving factory-default passwords active on a smart camera—human error and gaps in awareness remain the most targeted attack vectors. This underscores the critical value of security awareness training operating alongside technical controls. No single security tool provides absolute protection. A defence-in-depth strategy—layering complementary technical controls combined with proactive threat intelligence—is the only effective path forward.

3.2 Future Scope

To navigate an accelerating threat landscape, cybersecurity strategies must evolve across several fronts:

- **AI vs. AI Security:** As AI-powered attacks become ubiquitous, AI-driven defense—automated threat hunting, ML-driven anomaly detection, and real-time automated incident response—will become the baseline standard for enterprise Security Operations Centers (SOC).
- **Post-Quantum Cryptography (PQC):** Quantum computers capable of breaking current RSA and ECC encryption standards are projected within the next 10–15 years. Following NIST's finalization of the first post-quantum cryptographic standards in 2024, organizations must begin migration planning and establishing crypto-agility immediately.
- **Regulatory Evolution:** The implementation of the EU NIS2 Directive, strict SEC cybersecurity disclosure rules, and India's Digital Personal Data Protection (DPDP) Act

reflect a definitive global shift toward mandatory cyber incident reporting and board-level legal accountability.

- **Zero Trust as Default Baseline:** Moving beyond an elective framework, the US Federal Government's Executive Order signals that Zero Trust Architecture will become the standard baseline configuration across all public and private enterprise architectures globally.
- **Continuous Learning Imperative:** With threat actors rapidly adapting, cybersecurity education—encompassing advanced certifications, daily threat intelligence feeds, and hands-on lab environments—is an absolute necessity for modern IT professionals.

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